

Performance Testing

HDI Prediction System — Project Testing Phase

Testing Objective

Performance testing was carried out to verify that the HDI Prediction System returns accurate predictions reliably and responds quickly under normal usage, both locally and after deployment on Render.com.

Model Performance Testing

- Evaluation metric: R^2 (coefficient of determination), achieving approximately 0.98 on the trained dataset.
- Model tested against multiple sample countries with known HDI values to check prediction accuracy.
- Predicted HDI values found consistently close to actual published HDI figures.
- No significant overfitting observed given the strong linear relationship between features and target.

Functional & Load Test Cases

Test Case	Input Type	Expected Result	Status
Valid indicator values	All 4 fields filled correctly	Accurate HDI prediction returned	Pass
Missing field	One or more fields empty	Validation error shown	Pass
Non-numeric input	Text entered in numeric field	Input rejected with error	Pass
Extreme values	Very high/low indicator values	Prediction returned within reasonable range	Pass
Page load speed	Form and result pages	Loads within acceptable time	Pass
Deployed app availability	Public Render.com URL	Application accessible and responsive	Pass

Response Time Observations

- Local Flask server: prediction returned in under 1 second per request.
- Deployed Render.com instance: prediction returned within a few seconds, including network latency.
- No noticeable slowdown observed with repeated, successive form submissions.

Conclusion

The HDI Prediction System performs reliably across functional, accuracy, and response-time tests, confirming it is ready for demonstration and submission.